



A modern **full-stack e-commerce web application** built with **Node.js, Express.js, SQLite, and Vanilla JavaScript**.

MyShopNest delivers a complete online shopping experience with secure authentication, product browsing, cart management, checkout, payments, and admin controls.

Key Features

User Features

- Secure user registration & login
- Password hashing for enhanced security
- Browse products by categories
- Shopping cart functionality
- Seamless checkout process
- Razorpay payment integration
- View order history & order details
- Responsive mobile-friendly UI

Admin Features

- Admin dashboard access
- Product management
- Order management
- Secure admin-only routes

Technical Features

- SQLite database integration
 - RESTful API architecture
 - Environment variable configuration
 - Clean project structure
 - Local lightweight deployment
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Prerequisites

Before running the project, ensure you have:

- **Node.js** (v14 or higher)
 - **npm** (comes with Node.js)
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Installation & Setup

1 Clone the Repository

```
git clone <repository-url>  
cd MyShopNest
```

2 Install Dependencies

```
npm install
```

3 Configure Environment Variables

Create a `.env` file in the root directory:

```
PORT=3000  
RAZORPAY_KEY_ID=your_razorpay_key_id  
RAZORPAY_KEY_SECRET=your_razorpay_key_secret
```

4 Start the Application

Development Mode

```
npm run dev
```

Production Mode

```
npm start
```

Access the App

```
http://localhost:3000
```

Project Structure

```
MyShopNest/  
├─ backend/
```

```
| | | server.js
| | | db.js
| | | data/
| | |   | products.js
| | |
| | | frontend/
| | |   | index.html
| | |   | html/
| | |   |   | products.html
| | |   |   | cart.html
| | |   |   | checkout.html
| | |   |   | login.html
| | |   |   | register.html
| | |   |   | orders.html
| | |   |   | admin.html
| | |   | css/
| | |   | js/
| | |
| | | scripts/
| | |   | start-backend.js
| | |   | generate_project_report.py
| | |
| | | package.json
| | | README.md
```

Default Admin Credentials

 **For testing purposes only**

- **Email:** `admin@amazon-portfolio.local`
- **Password:** `Admin@12345`

 **Important:**

Change these credentials before deploying to production.

API Endpoints

Authentication

Method	Endpoint	Description
POST	<code>/api/auth/login</code>	User login

Method	Endpoint	Description
POST	<code>/api/auth/register</code>	User registration
POST	<code>/api/auth/logout</code>	User logout

Products

Method	Endpoint	Description
GET	<code>/api/products</code>	Get all products
GET	<code>/api/products/:id</code>	Get product details
GET	<code>/api/categories</code>	Get categories

Cart

Method	Endpoint	Description
GET	<code>/api/cart</code>	View cart
POST	<code>/api/cart/add</code>	Add item
DELETE	<code>/api/cart/:id</code>	Remove item
PUT	<code>/api/cart/:id</code>	Update quantity

Orders

Method	Endpoint	Description
GET	<code>/api/orders</code>	Get orders
POST	<code>/api/orders</code>	Create order
GET	<code>/api/orders/:id</code>	Order details

Payments

Method	Endpoint	Description
GET	<code>/api/payments/razorpay/config</code>	Razorpay config
POST	<code>/api/payments/razorpay/order</code>	Create payment order
POST	<code>/api/payments/razorpay/verify</code>	Verify payment

Admin

Method	Endpoint	Description
GET	<code>/api/admin/orders</code>	View all orders
GET	<code>/api/admin/products</code>	View all products
POST	<code>/api/admin/products</code>	Add product

Database Schema

Main Tables:

- **users** → User authentication data
- **products** → Product listings
- **categories** → Product categories
- **cart** → User shopping cart
- **orders** → Order records
- **order_items** → Items within each order

Application Workflow

```
User Registration/Login
    ↓
Browse Products
    ↓
Add to Cart
    ↓
Checkout
    ↓
Razorpay Payment
    ↓
Order Confirmation
    ↓
Order History
```

Security Features

- Password hashing with `crypto.scryptSync`
- Protected admin routes

- Environment variable security
 - CORS protection
 - Secure payment verification
-

Troubleshooting

Port Already in Use

Change PORT `in` `.env`

Reset Database

Delete `backend/data/store.db`

Razorpay Issues

GET `/api/payments/razorpay/config`

Ensure your credentials are correct.

Major Dependencies

- **express** → Backend framework
 - **cors** → Security middleware
 - **dotenv** → Environment configuration
 - **sqlite** → Database engine
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Future Enhancements

- [] Email notifications
 - [] Wishlist system
 - [] Product reviews & ratings
 - [] Advanced search filters
 - [] Multiple payment gateways
 - [] Inventory management
 - [] User profile customization
 - [] Real-time order tracking
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License

This project is licensed under the **MIT License**.

Author

Developed as a professional **e-commerce portfolio project**.

Support

If you encounter any issues or have suggestions:

Open an Issue on GitHub

Your feedback is always welcome.

 **If you like this project, consider giving it a star on GitHub!**

Happy Shopping! 